Perturbation Catalogue

An Integrated Resource for Exploring the Impact of Genetic Perturbations

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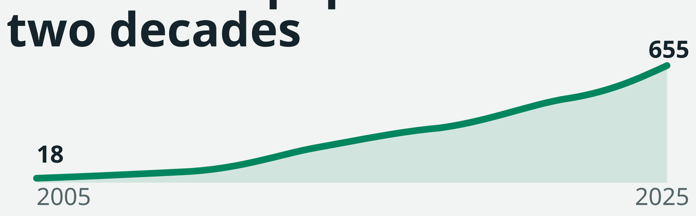
EMBL-EBI

Perturbation evidence is multiplying faster than it can be organised

A single genome-scale Perturb-Seq study can profile **more than 2.5 million cells**, while pooled screens test thousands of targets. That scale could reveal robust biology across studies—but identifiers, metadata, assays and processing pipelines differ. Results remain hard to compare or reproduce. The goal of the Perturbation Catalogue is to supply a shared schema, stable gene identity and one access path for combining this evidence.

36× growth in perturbation papers over two decades



Europe PMC title/abstract query; 24 Aug 2026.
Scale example: Replogle et al., Cell 2022.

Our approach turns scattered datasets into a unified resource

1,237 datasets | **19,235** human targets | **36** tissues | **34** cell types | **1,200** cell lines | **169** diseases

CRISPR screens
Pooled guide libraries perturb genes at scale. Phenotypes and fitness measurements reveal essentiality and context-specific dependencies.

Perturb-Seq
Guide identity is paired with single-cell RNA-seq, linking perturbations to cell states and gene-regulatory responses. Five reprocessed datasets are live.

MAVE
Variant libraries and functional assays measure sequence-to-function effects across many alleles, making subtle molecular changes searchable.

We extensively curate metadata...

84 meta-data fields

Study & provenance
Title, authors, year, source URI, licence, linked datasets

Perturbation & library
Targets, library design, delivery, guides, variants

Model system
Organism, tissue, cell type or line, disease, sex, stage

Readout & processing
Assay, technology, sequencing kit, reference, software

Stable Ensembl gene IDs and ontology-linked entities make the same biological question addressable across modalities.

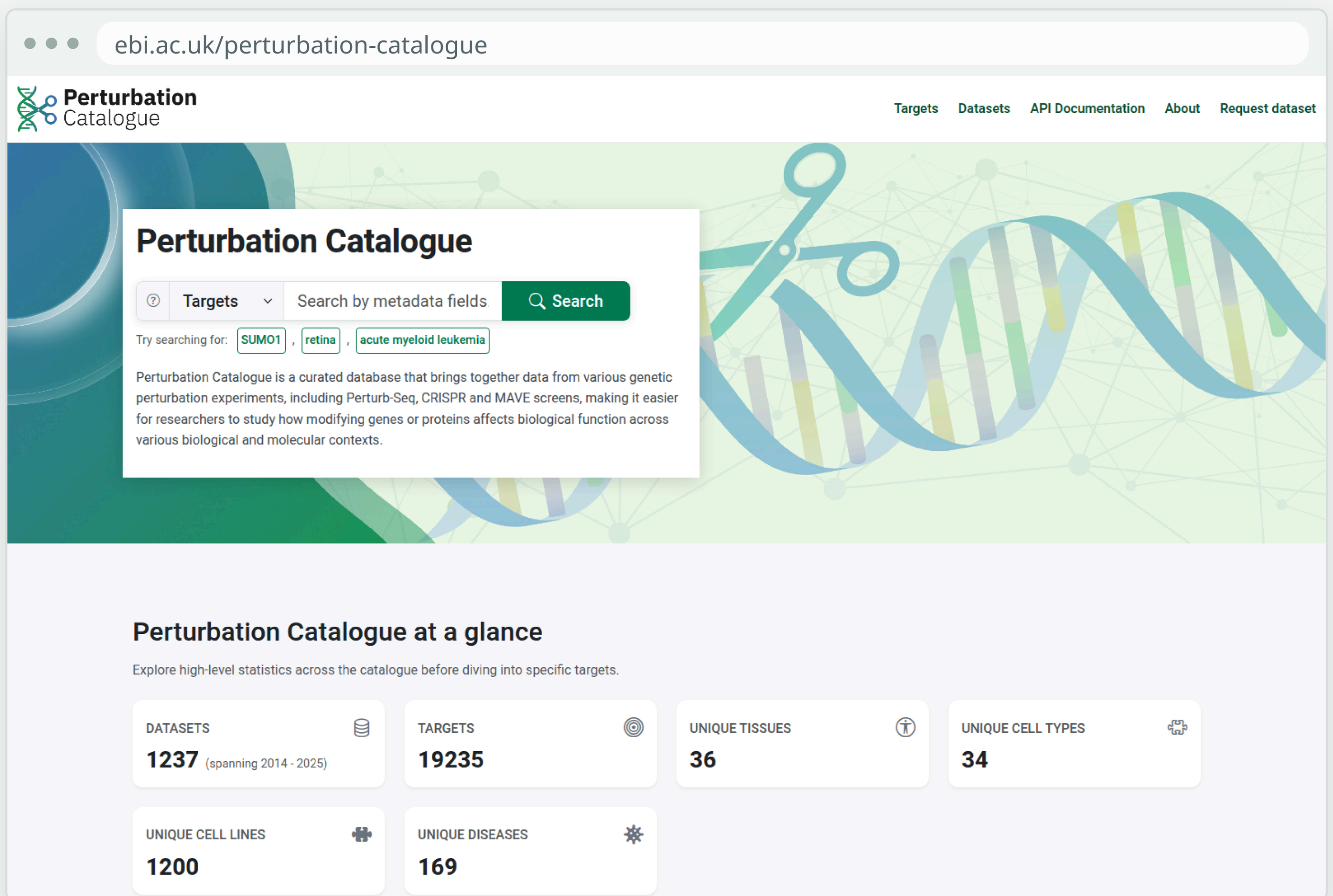
And reprocess Perturb-Seq data from scratch

Author matrices differ in alignment, annotation, filtering and guide assignment. Selected Perturb-Seq studies follow one shared computational path.

SRA / FASTQ — Raw sequencing reads and study-specific probe lists
→ **Cell × gene counts** — Shared reference and gene annotation
→ **Guide assignment** — Map cells to perturbations
→ **QC + filtering** — Produce the final analysis-ready H5AD

Five Perturb-Seq datasets are live. Perturbation–control comparisons feed DEA and Hallmark GSEA. Broad agreement with author-generated matrices is checked where available; independent quality metrics and wider kit support are next.

Browse the Catalogue



Target search, current coverage and biological-context summaries in one human-facing portal.

Access records via API

Search and retrieve metadata or modality results through the same backend as the portal.

```
GET /search?query=BRCA2&search_mode=targets
GET /v1/perturb-seq/nadig_2025_jurkat/search?limit=1
```


```
{
  "perturbation": {"gene_symbol": "TFAM"},
  "effect": {"gene_symbol": "MT-ND5", "log2fc": -1.926, "padj":
0.0}
}
```


Download whole datasets

Data: Parquet or CSV.gz. Metadata: JSON.

More exciting things are coming soon

Extract structured metadata faster while retaining curator review
AI-assisted drafting, ambiguity resolution and validated ontology mapping before ingestion.

Reanalyse more designs and measure quality independently
More kits and experimental structures, author-independent QC and modality-appropriate methods.

Publish periodic, versioned releases with explicit provenance
Stable snapshots and transparent analytical changes for reproducible downstream use.

Explore: **ebi.ac.uk/perturbation-catalogue** Get in touch: **perturbation-catalogue-help@ebi.ac.uk**

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